

KASA HARENDRA

Guntur, Andhra Pradesh, India | 6304340332 | [Linkedin](#) | harendrakasa@gmail.com | [Blog](#)

Computer Science undergraduate (GPA 9.28) with experience in high-performance computing, GPU-accelerated machine learning, and full-stack AI applications. Proven ability to design robust, modular software systems in Python, Java, and C.

SKILLS AND ABILITIES

- **Languages:** Python, C, Java, MySQL
- **Technical Skills:** Data Analysis using Python, Building Agents and Agentic workflows, Building RAG pipelines
- **Libraries/Frameworks:** PyTorch Pandas, Numpy, Scikit-Learn, Langchain, FastAPI, SQLAlchemy
- **Tools:** Git, Visual Studio Code, Jupyter Notebook, Ollama
- **Platforms:** Linux, Windows, Huggingface
- **Soft Skills:** Problem-Solving, Communication

EXPERIENCE

Intern – High Performance Computing (HPC)
Centre for Development of Advanced Computing (CDAC), Pune
April 2025 – June 2025

- Successfully completed a 2-month internship focused on the sub-project "**Graph Representation and Visualization of Telecom Traffic Data.**"
- Applied graph-based data structures and Python libraries (networkx) to represent and visualize complex telecom traffic datasets.
- Demonstrated inventiveness, strong problem-solving skills, and domain knowledge in HPC and data visualization.
- Gained hands-on experience with **GPU architecture and CUDA/OpenACC** for accelerating data visualization tasks.

PROJECTS

1. [OpenAgents](#) | [Agentic AI](#) | [Github](#)

- **TechStack:** FastAPI, Electron, React, LangChain, BrowserUse, Sqlite
- Developed a **Desktop AI assistant** featuring autonomous swarm of agents for Research, Browser, Terminal, File Management, and Integrations capable of executing complex user workflows across web, local system, email, and calendar services with about **90% accuracy**.
- Defined an autonomous flow generated dynamically by the Orchestrator agent based on the user's requirement.
- Enabled **per-agent LLM customization** and secure **Google OAuth authentication**, allowing users to orchestrate **specialized AI agents** powered by different local and cloud language models within a single application.

2. [GraphMind AI](#) | [GenAI](#) | [Github](#)

- **Tech Stack:** FastAPI, Next.js, React-Flow, MongoDB, Redis, LangChain, GraphRAG
- Built a **multimodal GraphRAG platform** that converts documents, web content, and video transcripts into knowledge graphs for advanced AI-powered search and reasoning.
- Developed user-friendly chat interface to ingest documents, inspect the graphs, entities and relations
- Engineered a custom **multi-tenant graph retrieval system** with N-hop traversal, asynchronous Celery processing, and Redis caching, enabling scalable context-aware retrieval while **reducing LLM hallucinations by 80%**.

3. [GitOracle](#) | [Full Stack AI](#) | [Github](#)

- **Tech Stack:** FastAPI, Next.js, MongoDB, Redis, Celery, TreeRAG
- Built a secure AI-powered GitHub workspace integrating GitHub OAuth that performs **repository-wide chat, PR summarization, and README generation, file-specific chat, file exploration** all without cloning repositories locally.
- Achieved **90%+ reduction in GitHub API calls** and **sub-second code retrieval and chat** using custom implemented heirarchial **TreeRAG-based indexing**, distributed Celery workers, and multi-layer caching for repositories with **thousands of files**.

4. [AI Data Tool \(Conversational Tabular Analysis\)](#) | [GenAI](#) | [Github](#)

- **Tech Stack:** Python, Streamlit, Pandas, LangChain (PandasAgent), Matplotlib, Seaborn, Plotly, PythonREPL.
- Built a conversational platform that empowers users to analyze CSV and Excel datasets through plain English. The tool automates statistical summarization, data cleaning, and complex visualization generation.
- Engineered a **transparent code execution** engine using **PythonREPL**, allowing for real-time validation and display of the generated Pandas logic.
- Optimized prompt engineering for **zero-inference accuracy**, ensuring the agent strictly adheres to the data context.

CERTIFICATIONS

- **CS50: Introduction to Programming with Python** – Harvard University
- **Generative AI Fundamentals** – Google Cloud Skill Boost
- **IBM Data Science Certificate** – IBM Cognitive Class
- **Competitive Programming** – NPTEL

Languages Known: English, Telugu, German(Intermediate)

EDUCATION

Vasireddy Venkatadri International Technological University
Bachelor of Technology (B.Tech); GPA: 9.28

Andhra Pradesh, India
Aug 2023 - Ongoing